Centripetal Force and Centripetal Acceleration

SYNOPSIS

Note 1 gave you the variables and the kinematics. This Note asks the question it deliberately left alone: if the velocity is turning, there is an acceleration, and therefore a force — how big are they, and which way do they point?

We derive the centripetal acceleration from the change in the velocity vector rather than asserting it. Two instants, the same speed, a small angle $\Delta\theta$ between them: the arc is $s = r\Delta\theta$, the speed is $v = s/\Delta t = r(\Delta\theta/\Delta t)$, and the change in the velocity vector has length $|\Delta\vec{v}| \approx v\Delta\theta$. Divide by Δt and the r falls out of the ratio: $a = v(\Delta\theta/\Delta t) = v(v/r) = v^2/r$. Then $F_c = ma_c$, and the job is always the same: find the real forces, add them, and set the sum equal to mv^2/r.

After that we let α stop being zero. The three rotational kinematics equations from Note 1's table come into their own, and the tangential-acceleration row that was dead weight there is suddenly the row you need.

We close with gravitation. $F = GM_1M_2/r^2$ is a centripetal force whenever an orbit is circular, and setting it equal to mv^2/r gives you the orbital speed, the orbital period, and — run backwards from g — the mass of the Earth.

BEFORE YOU READ ON — PREDICT

A roller coaster goes round a vertical loop. At the very top the track is *above* the car and the car does not fall off. Write one sentence saying which force holds it up there — and what happens to that force if the car goes slower.

PERFORMANCE EXPECTATIONS

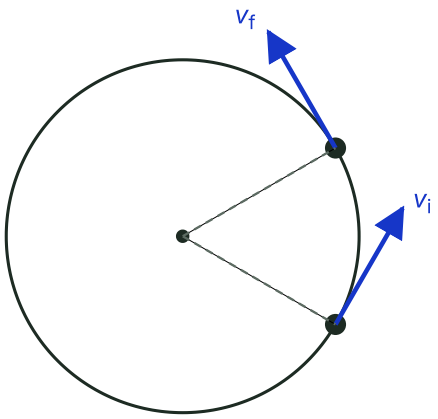
Derive the centripetal acceleration from the change in the velocity vector. Identify the real force that plays the centripetal role in a given situation and solve for speed, radius or force. Apply the rotational

kinematics equations when the angular acceleration is not zero. Use Newton's law of universal gravitation to find an orbital speed, an orbital period, or a planet's mass.

For PHY 151 — with calculus: Get $\vec{a} = -\omega^2\vec{r}$ by differentiating the position vector twice, so the direction of the centripetal acceleration falls out of a minus sign rather than a limit argument.

Centripetal acceleration, derived

Two instants a short time apart, the same speed at each. Redraw the two velocity vectors from a common tail and the difference between them is a short vector pointing sideways — inward. The triangle it makes is similar to the one the two radii make, and that similarity is the whole derivation.



two instants, a short time apart

Both velocity arrows are the same length — the speed never changed — and their difference is not zero.

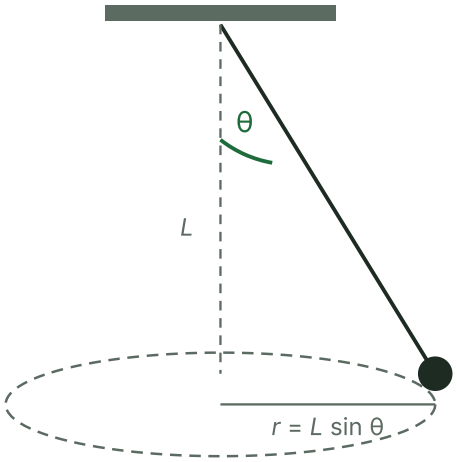
Written out, the way it goes in four lines:

arc length $s = r\Delta\theta$ speed $v = s/\Delta t = r(\Delta\theta/\Delta t)$, so $\Delta\theta/\Delta t = v/r$ Note 1's two equations, rearranged

$|\Delta\vec{v}| \approx v\Delta\theta$ the short side of an isosceles triangle with equal sides v and apex angle $\Delta\theta$

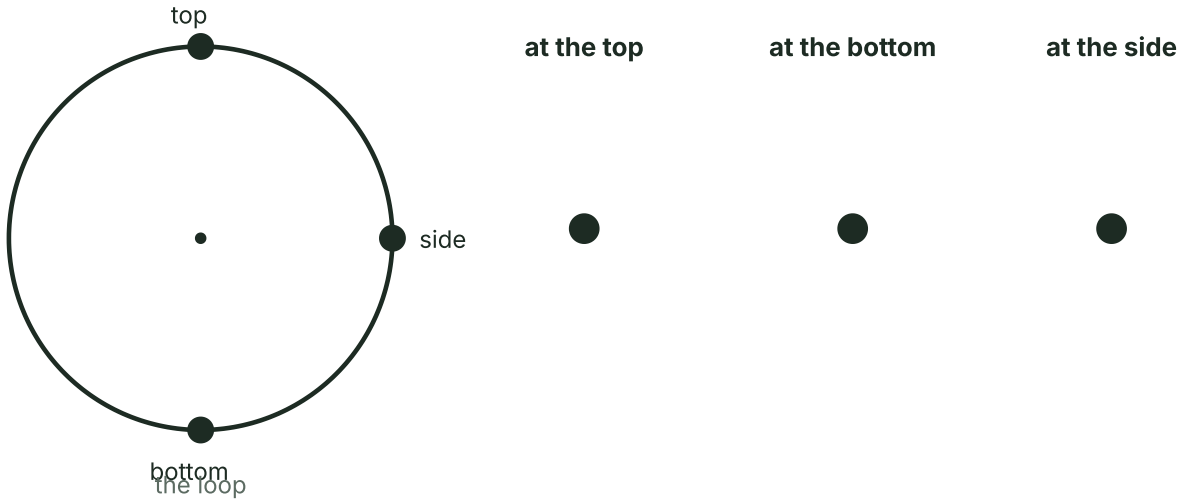
$a = |\Delta\vec{v}|/\Delta t = v(\Delta\theta/\Delta t) = v(v/r) = \boldsymbol{v^2/r}$ and it points where $\Delta\vec{v}$ points: at the centre

Centripetal is a job description, not a kind of force. You never draw an arrow labelled “centripetal force” on a free-body diagram. Draw the forces that are really there — tension, gravity, a normal force, friction — then notice that their sum points at the centre and equals mv^2/r.



A conical pendulum: the bob travels a horizontal circle, so the net force is horizontal and points at the axis. The vertical components must cancel, which is what fixes the tension.

1. A conical pendulum is a weight or bob fixed on the end of a string suspended from a pivot, moving in a horizontal circular path. (A) What is the tension in the string, in terms of m, g and θ? (B) What is the centripetal acceleration of the bob, in terms of g and θ?
2. The pendulum is 0.85 m long and swings at 18° from the vertical. (A) What is the magnitude of the centripetal acceleration, in m/s²? (B) What is the radius, in metres, of the horizontal circular path? (C) What is the speed of the mass, in m/s?



The same car at three places on the loop. The weight always points down; the normal force always points from the track toward the centre. What changes is whether they add or oppose.

3. Modern roller coasters have vertical loops. The radius of curvature is smaller at the top than on the sides, so the downward centripetal acceleration at the top is greater than g and the passengers stay pressed into their seats. What is the speed of the coaster at the top of the loop, in m/s, if the radius of curvature there is 17.0 m and the downward acceleration of the car is $1.50g$?
4. Write $F_c = mv^2/r$ in terms of F_N and F_g at each of the three places marked on the figure — top, bottom and side.
5. At the top of the loop, what is the **slowest** the car can go and still keep contact with the track? Work it symbolically.

Rotational motion with angular acceleration

Everything in Note 1 was uniform: $\alpha = 0$, so $a_t = r\alpha = 0$ and only the centripetal acceleration survived. Let the spin rate change and the three rotational kinematics equations do real work, and an object carries **both** accelerations at once, at right angles to each other.

$\omega = \omega_0 + \alpha t$ $\theta = \theta_0 + \omega_0 t + \tfrac{1}{2}\alpha t^2$ $\omega^2 = \omega_0^2 + 2\alpha\theta$ Module 1 in different letters, exactly as the table said

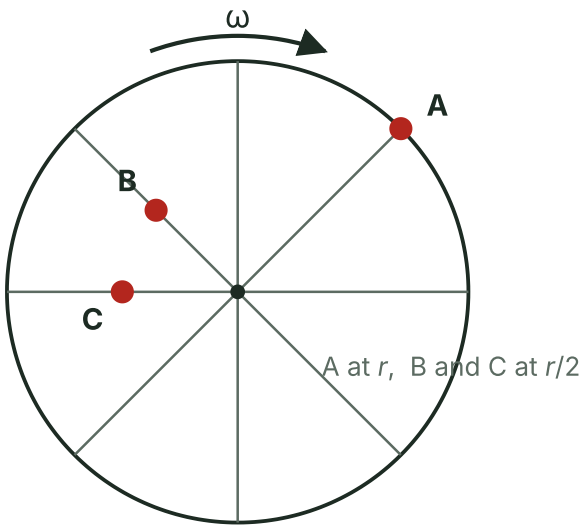
6. A track star runs a 295 m race on a 295 m circular track in 33 s. What is his angular velocity, in rad/s, assuming a constant speed?

7. On takeoff, the propellers on a UAV increase their angular velocity from rest at a rate of $\omega = 28.8t$ rad/s for 4.8 s. (A) What is the instantaneous angular velocity, in rad/s, at $t = 3.3$ s? (B) What is the angular acceleration, in rad/s²?

8. The angular velocity of a rotating rigid body increases from 450 to 1400 rev/min in 135 s. (A) What is the angular acceleration of the body, in rad/s²? (B) Through what angle, in rad, does it turn in this 135 s?

9. An ultracentrifuge accelerates from rest to 100 000 rpm in 2.50 min. (A) What is the average angular acceleration, in rad/s²? (B) What is the tangential acceleration, in m/s², of a point 4.50 cm from the axis of rotation? (C) What is the centripetal acceleration of that point at full rpm, in m/s² and in multiples of g? (D) What total distance, in metres, does that point travel?

Two numbers here differ from the ones in the printed handout. In 9(C) the handout's $4.93 \times 10^5\ g$ is a_c divided by 10 rather than by 9.80; the correct multiple is 5.04×10^5. In 9(D) the handout writes $\theta = 7.85 \times 10^3$ rad, but $\tfrac{1}{2}(69.8)(150)^2$ is 7.85×10^5 — an exponent slip only, because the distance that follows it, 35 336 m, is right.



A on the rim, B and C halfway out on two different spokes, over one interval Δt.

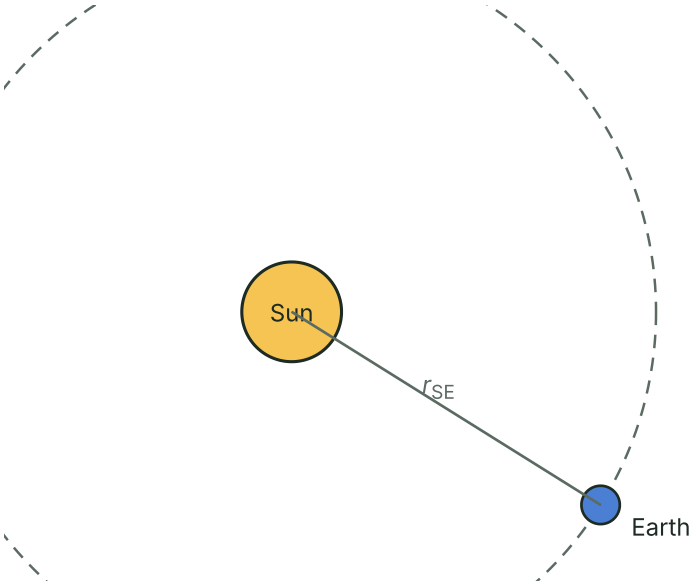
10. Which of the following is true regarding the **angular velocity**? (A) All three points have the same angular velocity, as all three have the same angular displacement θ in the same time interval Δt. (B) B and C have half the angular velocity of A, as they are at half the distance from the centre. (C) B and C have twice the angular velocity of A.
11. Which of the following is true regarding the **tangential speed**? (A) All three have the same tangential speed. (B) A has a greater tangential speed as it is further from the centre. (C) B and C have a greater tangential speed as they are closer to the centre.
12. Which of the following is true regarding the **centripetal acceleration**? (A) All three have the same centripetal acceleration. (B) B and C have half the centripetal acceleration of A. (C) B and C have twice the centripetal acceleration of A.

Universal gravitation

Every pair of masses attracts, along the line between them, with

$F_g = G\,M_1M_2 / r^2$, $G = 6.67 \times 10^{-11}\ \text{m}^3\cdot\text{kg}^{-1}\cdot\text{s}^{-2}$ and $F_{12} = -F_{21}$: a Newton's third law pair

When an orbit is circular, that force *is* the centripetal force — nothing else is acting. Setting the two expressions equal is the whole method, and the orbiting body's own mass cancels out of it.



and the third-law pair:



equal and opposite, along the line between them

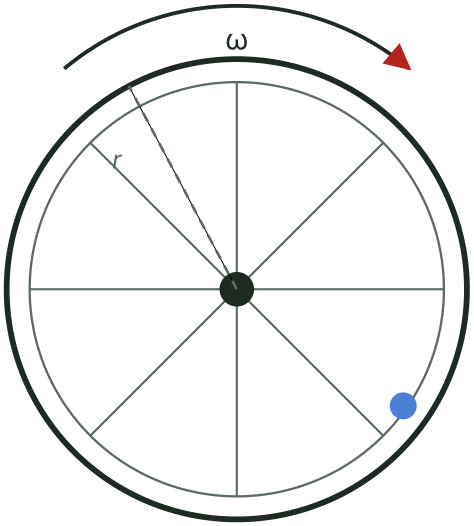
$F = GM_1M_2 / r^2$

The Earth in a circular orbit. Gravity is the only force on it, it points at the Sun, and it is therefore the centripetal force.

13. Set $F_c = F_g$ for the Earth's orbit and solve for the orbital speed v_E in terms of G, M_{Sun} and r_{SE}. Then use $v_E = 2\pi r_{SE}/T$ to get the orbital period T.

14. Evaluate T with $M_{Sun} = 1.989 \times 10^{30}$ kg and $r_{SE} = 150 \times 10^6$ km. How close is it to a year, and what does the discrepancy tell you?

15. Calculate Earth's mass given that the acceleration due to gravity at the North Pole is 9.830 m/s² and the mean radius of the Earth at the North Pole is 6371 km.



The floor is the OUTER wall.
It pushes the astronaut inward, toward the axis, and that push is the whole centripetal force.

Diameter 458 m, so r = 229 m.

A rotating space station. The floor is the *outer* wall, and the normal force it pushes inward with is the whole centripetal force — which is what the astronauts feel as weight.

16. A rotating space station is said to create "artificial gravity" — an acceleration crudely similar to gravity. The outer wall becomes the floor, and the centripetal acceleration supplied by the floor lets astronauts exercise and maintain muscle and bone strength more naturally. If the station is 458 m in **diameter**, what angular velocity, in rad/s, would produce an artificial gravity of 9.80 m/s² at the rim?

FOR PHY 151 — WITH CALCULUS

The derivation above is a limit argument with a similar-triangles step in the middle. With calculus it is two differentiations and a minus sign.

17. For uniform circular motion the position vector is $\vec{r}(t) = r\,(\cos \omega t)\,\hat{\imath} + r\,(\sin \omega t)\,\hat{\jmath}$. Differentiate twice. What does the result say about the magnitude and the direction of $\vec{a}$, with no limit argument at all?

18. Repeat with ω a function of time. A second term appears — identify it, and say which row of Note 1's comparison table it is.

Looking ahead

Centripetal force is Newton's second law with the geometry of a circle imposed on it; universal gravitation is the force law that makes orbits circular. **Note 3** changes the subject from a point going round a circle to a *body* turning about an axis, where mass that is far from the axis counts for more than mass close to it. That is the moment of inertia, and it is what the rest of the module is about.

Where this sits. Note 1 supplied θ, ω, α and the kinematics; this note supplied the dynamics. Between them they cover every circular-motion problem in the module that does not involve an extended body.

BACK TO YOUR PREDICTION

At the top of the loop, **gravity** is what holds the car in its circle — it points at the centre, which is exactly where the centripetal force has to point, and the track's push adds to it. Go slower and less centripetal force is needed, so the track pushes less; at $v = \sqrt{(gr)}$ the track has stopped pushing altogether and gravity alone is just enough. Slower than that and the car cannot stay on the track. If you answered “the track holds it up”, notice that the track pushes *down* at the top, not up.

Check yourself

1. A 1200 kg car rounds a flat curve of radius 45 m at 15 m/s. What centripetal force is needed, in N?

_____ N

2. At the very top of a vertical loop, a car just barely maintains contact with the track. At that instant

- ○ the normal force is equal to the car's weight
- ○ the normal force is zero and gravity supplies the whole centripetal force
- ○ the net force on the car is zero
- ○ the normal force points upward, away from the centre

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3. A satellite is moved to an orbit of four times the radius. Its orbital speed

- ○ is four times as large
- ○ is twice as large
- ○ is half as large
- ○ is one quarter as large

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4. A wheel starting from rest reaches 12.0 rad/s in 4.0 s. What is its angular acceleration, in rad/s²?

_____ rad/s²

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

print or open on a tablet; record your prediction, then work the items as the video goes through them.

AFTER THE MODULE

Completed notes (PDF)

released when Module 5 closes.

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